

Reader Report: Week 06

VI-BIP, ICP Theorems, & Causal Representation Learning

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Summary: Revisiting VI-BIP and the Core ICP Theorems

Wu and Blei (2025) frame ICP as sparse Bayesian variable selection. The binary selector $z \in \{0, 1\}^p$ marks invariant features; the posterior

$$p(z \mid \mathcal{D}) \propto p(z) \prod_{e,i} \frac{\hat{g}(Y_i^e \mid X_{ei}^z)}{\hat{p}_e(Y_i^e \mid X_{ei}^z)}$$

up-weights z whose pooled conditional $\hat{g}$ dominates each local conditional $\hat{p}_e$ —a likelihood-ratio test embedded in a prior. **Consistency (Thm 1)** requires only $\mu_{\min} > 0$ (positive KL-divergence between pooled and local conditionals on at least one bad feature); the posterior then concentrates on z^* . **Contraction (Thm 2)** sharpens this: the posterior mass on $\{z : z \neq z^*\}$ decays as $O(R e^{-\kappa n E \mu_{\min}})$, where $R = 2^p$ is the model count, E the number of environments, and κ a universal constant. Two messages emerge: (i) the exponential in $n E \mu_{\min}$ means *either* more data or more diverse environments can substitute for each other; (ii) because $R = 2^p$ appears only in the pre-exponential, large feature spaces slow convergence polynomially in p , not exponentially—a surprisingly benign curse of dimensionality.

VI-BIP approximates $p(z \mid \mathcal{D})$ via a mean-field family $q(z) = \prod_j q_j(z_j)$, optimizing ELBO. The obstacle is that the ELBO gradient involves discrete z ; the paper applies the **U2G estimator** (unbiased-to-Gaussian), which reparameterizes the discrete draw through a continuous Gaussian relaxation before taking gradients. This allows standard stochastic gradient ascent despite the combinatorial support. The approximation cost is a gap between the true posterior and q^* ; in practice, on a 6,170-gene perturbation dataset, VI-BIP recovers the same invariant genes as exact BIP while being orders of magnitude faster. Fan et al. (2024) offer a complementary perspective: recast ICP as a continuous minimax problem

$$\min_b \max_{w \in U(\gamma)} \sum_e w_e \hat{\mathbb{E}}[(Y^e - b^\top X^e)^2],$$

where $U(\gamma)$ admits negative weights. By contrast with BIP’s combinatorial posterior, NegDRO has a convex inner problem and differentiable outer problem, enabling polynomial-time convergence. The conceptual gap is that BIP produces a *distribution* over invariant sets—revealing ambiguity when environments are few—while NegDRO returns a single point estimate $\hat{\beta}^\gamma$ with explicit finite-sample rates but no uncertainty quantification.

Causal Representation Learning (Schölkopf et al.)

Schölkopf et al. (2021) argue that robustness and out-of-distribution generalization require not merely statistical patterns but an *approximate causal model* of the data-generating process. Their central organizing principle is the **independent causal mechanisms (ICM)** conjecture: the causal Markov factorization $p(X_1, \dots, X_d) = \prod_i p(X_i \mid \text{pa}_i)$ decomposes reality into modules that are *independently* modifiable by interventions. Anti-causal directions—modeling effects given causes—violate ICM because they conflate mechanism with marginal, yielding fragile predictors that break under distributional shift. This connects directly to ICP: Peters et al.’s invariant set S^* is precisely the set whose *causal* conditional $p(Y \mid X_{S^*})$ is stable, while anti-causal features’ conditionals shift.

The paper surveys three operationalizations of ICM: (1) independence of noise variables in additive SEMs (link to ICP); (2) Kolmogorov-complexity arguments implying that $p(\text{cause})$ and $p(\text{effect} \mid \text{cause})$

compress independently; (3) disentangled representation learning, where latent variables recover causal factors. While the survey is technically rich, it does not yet deliver a single unified algorithm—the “messiness” Blei notes is real: the paper is a call to arms rather than a solved problem.

Critique & Project Connection

For my *Predicting NIMBYism Causally* project the Schölkopf framework clarifies a modeling choice I had left implicit. My protest indicator Y (zoning opposition) is an *effect* of regulatory and demographic causes; I have been modeling $p(Y | X)$, an anti-causal direction, which ICM theory warns will be fragile across the policy regimes I use as environments. The appropriate posture is to encode the regulatory shifts as *interventions* on the causal graph and identify S^* as the features whose *causal* conditional is intervention-stable—precisely what ICP formalizes. VI-BIP’s posterior is especially useful here: with only four temporal environments and ~ 50 features, the posterior over z will be multi-modal, and those multi-modal regions identify the features where additional environment diversity is most needed. NegDRO’s fast point estimate can serve as a computational screen before committing to the full VI-BIP run.

Questions

1. The contraction rate $O(R e^{-\kappa n E \mu_{\min}})$ treats all $R = 2^p$ candidate sets equally in the pre-factor. In practice, sparsity priors (e.g. Beta-Bernoulli with small mean) reduce the effective model count. Does the contraction result sharpen when $p(z)$ concentrates on sparse z , or does the 2^p bound remain tight regardless of the prior?
2. Schölkopf et al. argue that anti-causal predictors are less robust, yet NegDRO optimizes a squared-loss objective on $p(Y | X)$ —ostensibly anti-causal. NegDRO’s invariance guarantee holds under additive interventions on X ; does this suffice to recover causal features even when the regression direction is anti-causal, or does an anti-causal orientation introduce a systematic bias that its finite-sample rates do not capture?

References

- Y. Fan, S. Li, and Z. Yang. Causal invariance learning via efficient nonconvex optimization. *arXiv preprint arXiv:2412.11850*, 2024. URL <https://arxiv.org/abs/2412.11850>.
- B. Schölkopf, F. Locatello, S. Bauer, N. R. Ke, N. Kalchbrenner, A. Goyal, and Y. Bengio. Toward causal representation learning. *Proceedings of the IEEE*, 109(5):612–634, 2021. doi: 10.1109/JPROC.2021.3058954.
- Y. Wu and D. M. Blei. Bayesian invariance modeling of multi-environment data. *arXiv preprint arXiv:2506.22675*, 2025. URL <https://arxiv.org/abs/2506.22675>.